

Human-Computer Interaction for the Mass Customization of 3D-Printed Eyewear: Designing an Aesthetically Fitting System and Its Co-creation Services

Ding Zhou*
Queensland College of Art and
Design
Griffith University
Brisbane, Australia
School of System Design and
Intelligent Manufacturing
Southern University of Science and
Technology
Shenzhen, China
zhoud3@sustech.edu.cn

Erdong Xing
School of System Design and
Intelligent Manufacturing
Southern University of Science and
Technology
Shenzhen, China
1528045870@qq.com

Jiawei Huang
School of System Design and
Intelligent Manufacturing
Southern University of Science and
Technology
Shenzhen, China
12431366@mail.sustech.edu.cn

Abstract

This project explores the integration of mass customization and 3D printing in eyewear design to develop an aesthetically fitting system that harmonizes eyeglass frames with individual facial features. Utilizing human-computer interaction, the system leverages user facial data for personalized frame selection and adjustments, addressing significant knowledge gaps in the current market. It designs a co-creation service to enhance user engagement throughout the customization process, emphasizing the importance of user feedback in driving design decisions. Key lessons learned include the critical role of aesthetical experience in user choice, the need for intuitive navigation to mitigate choice overload, and the value of iterative design through user interaction. The project highlights the potential of combining traditional eyewear design with modern technology to create personalized, functional, and aesthetically pleasing products. In this way, it paves the way for future research in integrating artificial intelligence and machine learning for even more refined customization experiences.

CCS Concepts

• Human-centered computing; • Interaction design; • Interaction design process and methods; • Activity centered design;

Keywords

Human-computer interaction, Mass customization, Aesthetically fitting system, Co-creation service

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*Corresponding author

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1 Introduction

The eyewear industry has adopted mass customization, enabled by additive manufacturing/3D printing, to streamline the computational design of eyewear frames, ensuring aesthetical harmony with individual facial features. Mass customization leverages modern manufacturing technologies to meet diverse and highly personalized consumer needs efficiently [1]. Compared to traditional production methods, this approach more rapidly and accurately satisfies consumer demands while improving production efficiency and resource utilization [2]. A mass customization model tailors product design and production to consumer needs. Consumers can select styles, sizes, colors, and materials within a given range, and personalize products with logos, patterns, or text. This approach not only delivers more satisfying products and services but also increases customer loyalty and brand value. Moreover, it improves efficiency, lowers costs, and enhances production line flexibility, giving companies a competitive edge. However, a fully optimized aesthetical fitting system for eyewear that fully leverages 3D printing in mass customization remains underexplored—**Knowledge Gap 1**.

Chinese brands like BRAGi have developed 3D-printed customized eyewear, offering personalized eyeglass frames that fit various face sizes and shapes, resulting in a more comfortable and precise visual experience [3]. The production process involves seven steps: style selection, facial scanning, wear simulation, design generation, 3D printing, post-processing and dyeing, and delivery. Facial scanning technology is used to capture user data, followed by 3D modeling software to design the eyewear frame. The frame is then manufactured using 3D printers, and lenses are added to complete the glasses. This process exemplifies co-creation, as it transforms consumers into prosumers, with 3D printing technology fully enabling the potential of this collaborative approach [4]. In a business strategy centered on customer experience and interactive

relationships, co-creation refers to an active, creative, and social collaborative process between producers and users, aimed at generating value for customers [5]. From an industrial perspective, co-creation is thus expected to achieve a rapidly growing market volume [6, 7]. However, a co-creation service for consumers with mass customization of 3D printed eyewear has yet to be fully explored—**Knowledge Gap 2**.

In the context of Human-Computer Interaction (HCI), this project tackles the challenge of achieving aesthetic harmony between 3D-printed customized eyewear and users, aiming to create an aesthetically fitting system for eyeglass frames based on individual facial data. It also seeks to develop an interactive service model aligned with the co-creation process. This approach addresses the aesthetical and personalized fitting limitations of current 3D-printed eyewear while promoting a deeper integration of facial data into the design process, thus expanding both the theoretical and practical frontiers of customized design. The project provides new methodological support for personalized product design and optimizes the customization production process, offering significant theoretical value and broad application prospects.

2 Related Work and Project Purpose

In addressing Knowledge Gap 1, while existing practices and research have enhanced the aesthetical harmony between eyewear and facial features, an aesthetically fitting system fully driven by individual facial data remains unrealized. The KAZAK eyewear brand has developed an intelligent system that integrates machine learning and artificial intelligence to analyze facial features, face shape, hairstyle, and skin tone, generating a personalized aesthetic score to guide users in selecting the most suitable eyewear style. Users simply upload a facial photo, and the system uses algorithms to assign an aesthetic score from 1 to 5, with 5 representing the optimal style [8]. The technical approach of the system comprises four stages: data collection, data processing and analysis, feature engineering, and machine learning. This process involves collecting and anonymizing facial data to protect privacy while extracting facial features through standardization, normalization, and dimensionality reduction. Machine learning algorithms are then applied to analyze these features and predict aesthetic scores, enhancing the accuracy of user choices. In addition, Maruoka et al. proposed an intelligent system that utilizes an interactive genetic algorithm to merge user aesthetic preferences with suitability data for optimizing eyewear design [9]. This system generates recommendations based on user evaluations, using frame shape and color as algorithmic chromosomes. Frame shape is represented by 3 bits, while color is expressed using hue, saturation, and value (HSV), represented by 12 bits. The 15-bit chromosome can express up to 32,768 different design options. The study validated the system's effectiveness in optimizing eyewear design through experiments comparing genetic algorithm versions based on nonlinear transformation and individual roulette methods. Results indicated that the individual roulette method yielded better optimization, significantly enhancing user satisfaction [9]. However, the aforementioned practices and research have not fully leveraged the advantages of mass customization through 3D printing, failing to simultaneously achieve the alignment of eyeglass frames with various face shapes and

the aesthetical adjustment to individual facial features—**Research Problem 1**.

In addressing Knowledge Gap 2, while current customization of 3D-printed eyewear primarily relies on digital interaction to gather facial data and provide feedback during the customization process, a co-creation service model in the context of an aesthetically fitting system remains unrealized. A trust model should serve as the foundation for interaction in the intelligent frame customization system. Similar recommendation systems are often based on trust models [10, 11], which encompass system functionality, agent trustworthiness, and trust intention. System functionality pertains to design aspects such as interface display technologies, algorithms, and user interaction models. Agent trustworthiness refers to users' perceptions of the system's competence and reputation, directly influencing their trust intentions. This project aims to analyze how Chinese users perceive system capabilities and how these perceptions affect their trust. Trust intention includes the expected benefits users seek, such as the desire to purchase the designed 3D-printed eyewear or to access additional recommendation information. It also emphasizes the role of trust in reducing users' cognitive load. In addition, this research explores methods for establishing trust through interface display technologies, covering content selection, and the generation of explanations [12]. It also examines how different explanation formats, such as graphics and text, and the balance of information, affect user trust. The interaction experience of the frame customization system should comprehensively consider effectiveness, efficiency, and satisfaction. Effectiveness and efficiency are common evaluation parameters; however, 'satisfaction', mainly overlooked, reflects users' subjective experiences, including pleasure and comfort. Therefore, this project emphasizes these concepts in the customization service for 3D-printed eyewear. Research indicates that interaction design frequently neglects the 'darker side' of visual, aesthetical, and emotional aspects. To enhance communication among designers, scholars, and practitioners, interface design should not only improve usability but also emphasize aesthetics. This shift from a rational to an emotional focus reinstates the core role of visual design. Tractinsky et al. [13] and Norman [14] provide important references for improving system effectiveness, efficiency, and satisfaction. However, they have yet to address the customization service for aesthetically fitting 3D-printed eyewear and user experience, and have not established an appropriate co-creation service model—**Research Problem 2**.

Furthermore, Research Problems 1 and 2 guide the two **Objectives** of this project, respectively:

- Designing and developing an aesthetically fitting system that can (i) match various types of eyeglass frames to different face shapes and (ii) adjust some parts of an eyeglass frame for individual facial harmony.
- Designing and developing a co-creation service to apply the aesthetically fitting system to provide consumers with mass customization of 3D-printed eyewear.

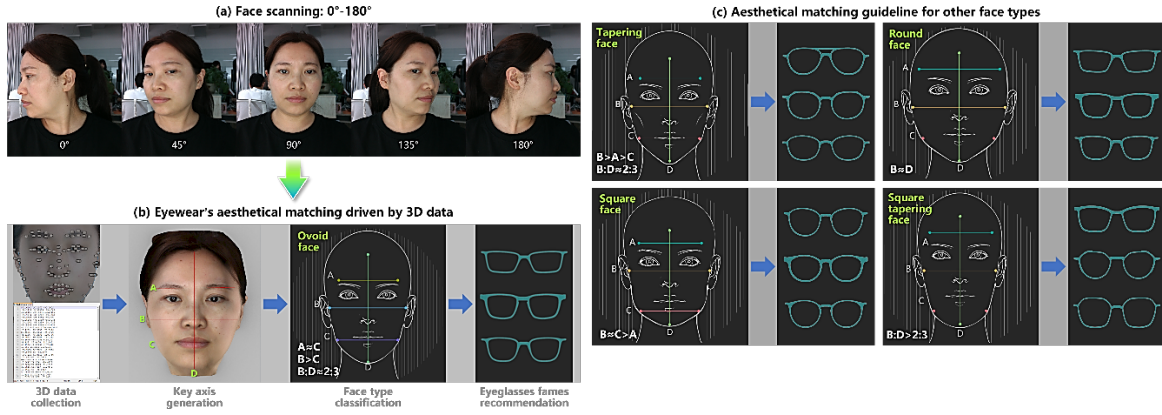


Figure 1: The Principle and Rules of Aesthetical Matching of Eyeglass Frames to Different Face Types.

3 Design Methods

3.1 Designing an aesthetical fitting system for the mass customization of 3D-printed eyewear

3.1.1 *Aesthetical matching of eyeglass frames to different face types.* To achieve **Objective 1 (i)**, we established the principles and rules for the aesthetical matching of eyeglass frames to different face types (see Figure 1):

- The user begins with a digital facial scan in a well-lit environment, where facial images are captured from five different angles: 0 degrees, 45 degrees, 90 degrees, 135 degrees, and 180 degrees. This scan uses an image acquisition device to obtain clear and comprehensive facial images.
- These images generate a 3D facial model, followed by feature point recognition to extract the 3D coordinates of key feature points. After defining these key points, the system automatically divides the face into three regions, i.e., upper, middle, and lower, based on the 3D coordinates of the feature points and preset guidelines. In this way, it generates four key facial axes: A) distance between the outermost points of the two eyebrows, B) inter-cheekbone distance, C) inter-jaw distance, and D) distance from the hairline to the jawline. Then, it determines a facial shape based on the proportional relationships of these axes, providing appropriate frame recommendations. For example, when $A \approx C$, $B > C$, and $B:D \approx 2:3$, the face is classified as an ovoid face, and three specific frame styles are recommended to enhance overall facial harmony.
- Aesthetical matching guidelines for other facial types include: when $B > A > C$ and $B:D \approx 2:3$, the face is classified as a tapering face, with three specific frame styles recommended to emphasize the facial curves. When $B \approx D$, the face is categorized as a round face, and three specific frame styles are endorsed to soften the round features. When $B \approx C$, the face is sorted as a square face, and three specific frame styles are suggested to soften the angularity of the face. When $B:D > 2:3$, the face is classified as a square tapering face, and three specific frame styles are recommended to reduce the elongated appearance of the face.

3.1.2 *Aesthetical adjustment of eyeglass frames for individual facial harmony.* To accomplish **Objective 1 (ii)**, we developed aesthetical adjustment rules for four different parts of eyeglass frames: upper rim position, lower rim position, middle beam height, and middle beam width (see Figure 2).

Table 1 represents the aesthetical adjustment rules for eyeglass frames, designed to promote individual facial harmony.

3.2 Designing a co-creation service based on the aesthetical fitting system

To reach **Objective 2**, we adopted the ‘Building Blocks of Interactions for Value Co-creation’ proposed by Prahalad and Ramaswamy [15] as the theoretical foundation, systematically observing and describing qualitative data. This model includes Dialogue, Access, Risk-benefits, and Transparency (DART), which are essential components of the co-creation interactions between consumers and service providers. The DART model requires service providers to offer access to product information and to achieve transparency in service through dialogue during the customer co-creation process. Subsequently, access and transparency enable customers to identify the risks and benefits of the service [16]. As shown in Figure 3, Zhou et al. [1] explored the mechanisms of online 3D design applications in-depth, and their findings further refined the ‘Design Model of the Experience of Co-creation with 3D Printing’.

The first two access points guide customers through two rounds of dialogue. Dialogue 1 introduces the services, while Dialogue 2 allows customers to co-create products. In Dialogue 3, customers have two options for obtaining the co-created product: downloading or purchasing. The first option enables customers to download and print the final product themselves, while the second directs them to purchase co-created products, which are produced by the service provider in an opaque manner. This design model provides a framework for creating a co-creation service based on the aesthetical fitting system for the mass customization of 3D-printed eyewear.

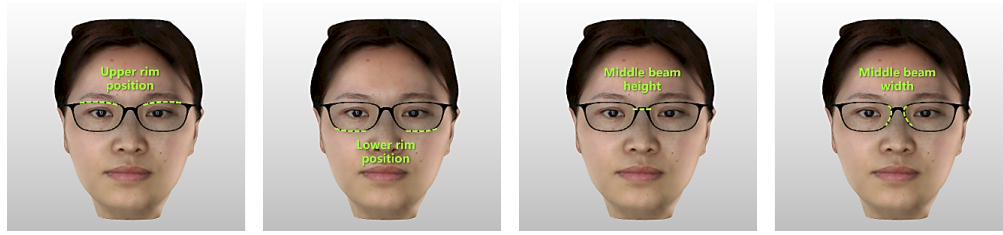


Figure 2: Different Parts of Eyeglass Frames in Aesthetical Adjustment for Individual Facial Harmony.

Table 1: Aesthetical adjustment rules of eyeglass frames for individual facial harmony

Parts of Eyeglass Frames	Aesthetical Adjustment Details
Upper rim position	- The upper rim is positioned slightly above the eye-brow line, with the distance between the brow and the upper rim being $1/5$ to $2/5$ of the distance from the brow to the eyes. - If the upper facial region length exceeds half the total eyeglass frame height, the upper region is considered too long. In such cases, the frame model will automatically adjust the height or length of the frame to visually shorten the upper facial region. - If the distance between the brow and eyes is smaller than the width of the eyes, it is considered narrow. The frame model reduces the frame width to ensure that at least $3/5$ of the space remains between the brow and the eyes, alleviating the sense of crowding in the eye area.
Lower rim position	- The lower rim is positioned in the middle of the atrium. The distance from the eyes accounts for $2/5$ to $3/5$ of the atrium position. - If the height of the middle facial region exceeds $3/4$ of the user's nose width, the middle facial region is considered too long. The lower rim position will exceed the preset range, causing the frame model to lower the lower rim, which increases the distance ratio between the lower rim and the eyes, making the middle facial region appear shorter.
Middle beam height	If the height of the middle facial region is less than two-thirds of the user's nose bridge width, the ratio of the middle facial region is considered too short. The middle beam height is set between 0.1 and 0.2 times the length of the middle facial region; a higher middle beam height can make the middle facial region appear longer.
Middle beam width	- If the distance between the eyes occupies more than one-third of the facial width, the eye distance is large. The middle beam width is set between 1.2 and 1.5 times the eye distance; a larger middle beam width can make the eye distance appear shorter. - If the width of the nose bridge occupies more than one-quarter of the facial width, the nose bridge is considered wide. The middle beam width is set between 1.1 and 1.3 times the width of the nose bridge; a slightly larger middle beam width can make the nose bridge appear narrower. - If the nasal ala width occupies more than one-half of the facial width, the nasal ala appears wide. The middle beam width is set between 1.1 and 1.4 times the nasal ala width; a slightly larger middle beam width can make the nose less prominent.

4 Design Results

4.1 An aesthetical fitting system for the mass customization of 3D-printed eyewear

4.1.1 Parametric modeling for aesthetical matching of eyeglass frames to different face types. To achieve **Objective 1 (i)**, we successfully designed and developed components of the aesthetically fitting

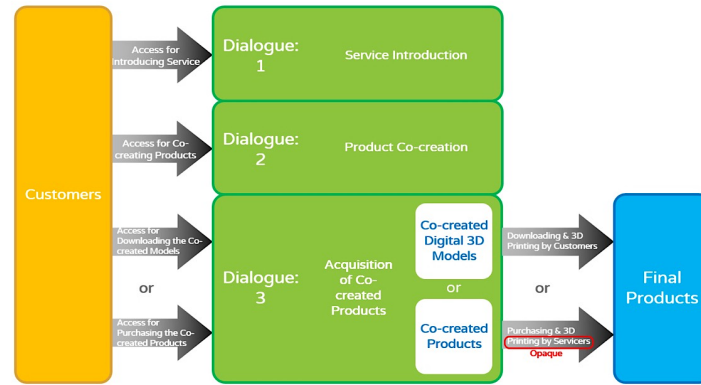


Figure 3: Design Model of the Experience of Co-creation with 3D Printing [1].

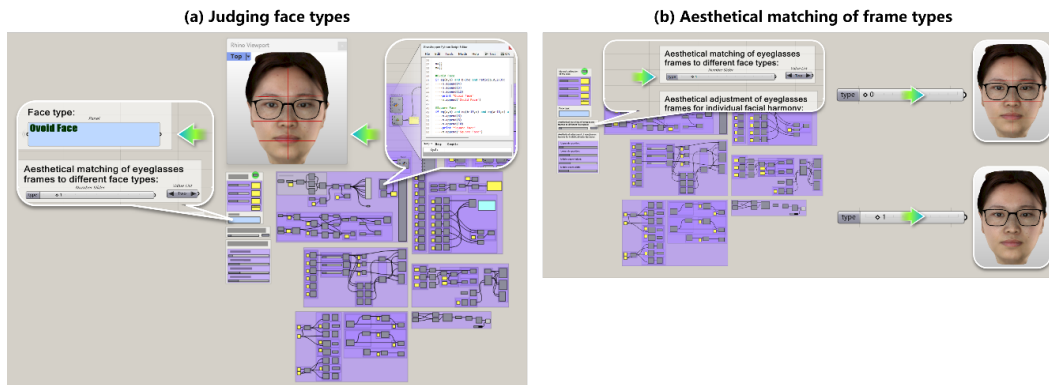


Figure 4: Parametric Modeling for Aesthetical Matching of Eyeglass Frames.

system to match various eyeglass frames to different face shapes. As shown in Figure 4 (a), the system utilizes the parametric design plugin *Grasshopper* within *Rhinoceros 3D*. Using Python code, it calculates and analyzes the proportional relationships of four generated auxiliary lines, allowing the code to autonomously determine and output the user's face shape information. Furthermore, this *Grasshopper*-based system can recommend different frame types based on facial shapes. In Figure 4 (b), users can manually adjust digital sliders to switch between recommended frame styles. Together, these processes fully implement the principles and rules of aesthetical matching of eyeglass frames to various face types, regarding Figure 1.

4.1.2 Parametric modeling for aesthetical adjustment of eyeglass frames for individual facial harmony. To reach **Objective 1 (ii)**, we successfully designed and developed components of the aesthetically fitting system, allowing for adjustments to specific parts of eyeglass frames to achieve individual facial harmony. As shown in Figure 5, after determining the frame type, the local shape of the frame model is adjustable according to the user's facial features by manually sliding digital sliders. This process facilitates aesthetic harmony. In this *Grasshopper*-based system, the adjustable parts

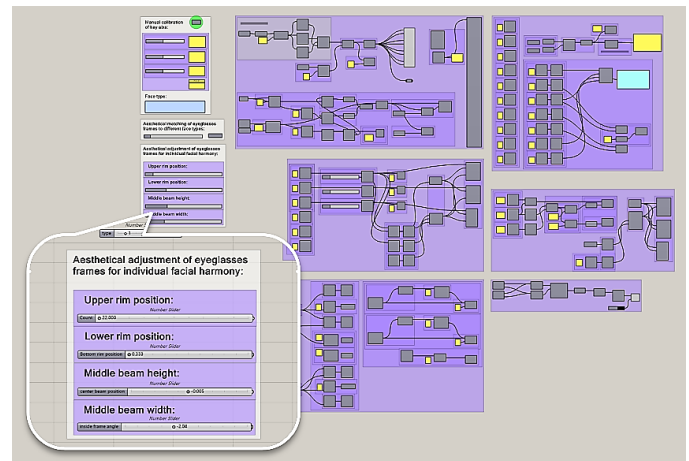


Figure 5: Parametric Modeling for Aesthetical Adjustment of Eyeglass Frames for Individual Facial Harmony.

of the eyeglass frames include the upper rim position, lower rim position, middle beam height, and middle beam width.

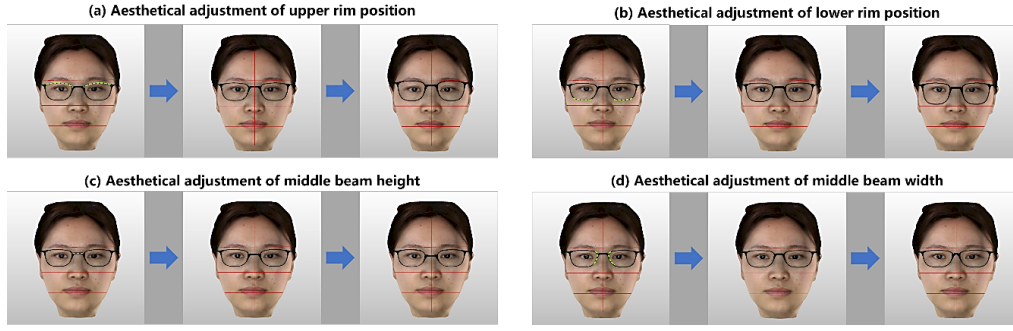


Figure 6: Aesthetical Adjustment of Different Parts of Eyeglass Frames.

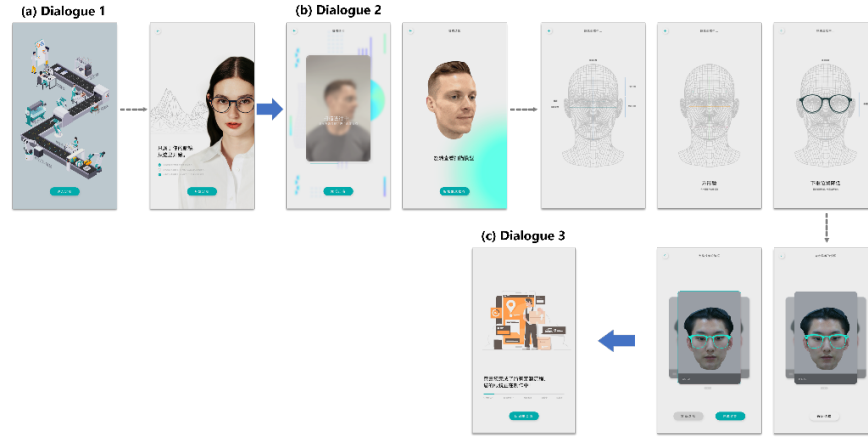


Figure 7: Interactive User Interface of the Co-creation Service Based on the Aesthetical Fitting System

Figure 6 specifically illustrates the process of adjusting the local shapes of the frame models to achieve aesthetical harmony. The comparisons in (a), (b), (c), and (d) demonstrate the different aesthetical effects produced by varying local adjustments. Consequently, these processes fully implement the aesthetical adjustment rules for eyeglass frames aimed at achieving individual facial harmony, as outlined in Table 1.

4.2 A co-creation service based on the aesthetical fitting system and its evaluation results

To fulfill **Objective 2**, we successfully designed and developed a co-creation service that utilizes the aesthetically fitting system to offer consumers mass customization of 3D-printed eyewear. Based on the design model of the co-creation experience with 3D printing demonstrated in Figure 3, we collaborated with our industry partner, BRAGi Co. Ltd., to automate all manual operations outlined in Section 4.1. In addition, we establish an interactive user interface for the co-creation service using the aesthetically fitting system (see Figure 7).

In Figure 7 (a), Dialogue 1, customers can access an introduction to the service, starting with a first page that provides an overview of the 3D printing customization process for eyewear. This page

features an interactive illustration of a conveyor belt displaying the production workflow once the custom order is completed. The workflow includes 3D modeling, 3D printing, polishing, dyeing and drying, lens cutting and assembly, quality inspection, and packaging. The second page serves as the starting interface, which includes reading general terms, checking agreements, and noting special conditions, such as astigmatism over 300 degrees or myopia exceeding 600 degrees. In Figure 7 (b), Dialogue 2, customers can participate in product co-creation. Initially, the system displays the facial scanning interface, consisting of two pages. On the first page, users will conduct facial scans under voice prompts and guidance from staff. After the scanning process, the system switches to the second page, displaying the facial scanning model for user verification. Then, the system plays an animated demonstration of computational aesthetical customization, showcasing the process of customizing eyeglass frames. This process includes matching different types of eyeglass frames to the user's face shape and adjusting the four parts of the frame for the user's facial harmony. During this process, the system backend analyzes the user's facial scan and feature data, generating a frame for the user based on the computational aesthetical customization logic. Finally, the system displays the frame generation and selection page, presenting three customized eyeglass options for the user to choose from. The user must confirm their selection to finalize the chosen frame. In Figure

7 (c), Dialogue 3, customers can access the co-created product. The system displays the completion page for customization, informing the user of the current status of their order. At this point, the user's order has been sent to the brand's backend for preparation, completing the eyewear customization process. The user can either wait for the finished glasses to be shipped home or pick them up in-store once they are ready.

5 Lessons Learned and Future Work

In the personalized customization process of 3D-printed eyewear, we have deeply applied the HCI principles and gained significant insights. First, user feedback has proven to be a critical driving force in design decision-making. Through multiple interactions with users during the early stages, we gathered their aesthetic and functional expectations while encouraging their active participation in the design process. This engagement enhanced their sense of identification and satisfaction with the final product. The iterative design approach quickly identifies and addresses issues within the design, improving the product's usability and appeal. Second, we recognized the importance of aesthetic experience in the user selection process. As a personalized accessory, eyewear should not only meet functional requirements but also align with users' personal styles and aesthetic preferences. Accordingly, we employed a parametric design approach, deconstructing the product into controllable factors within *Grasshopper*, driven by the user's facial data to achieve aesthetic harmony. This method captures diverse aesthetic needs and integrates them into the parametric product design, resulting in eyewear that fulfills practical requirements while maintaining a high degree of personalization. We also noted that users might experience confusion due to the complexity of choices during the customization process. This insight emphasizes the necessity of providing intuitive navigation and clear guidance when designing the interactive interface to facilitate a smooth customization experience. To alleviate the decision-making burden on users, we designed a co-creation service model including three dialogues that guide users through the service introduction, participation in product co-creation, and acquisition of the co-created product. This model not only enhances the user experience but also streamlines the customization process, ensuring that users feel supported and empowered throughout their journey. Therefore, these findings indicate the strength of employing the HCI principles to provide an engaging, user-centric customization experience for 3D-printed eyewear, enhancing a close relationship between users and their personalized products.

Furthermore, our project opens several avenues for future research. It plans to explore the integration of artificial intelligence and machine learning technologies into the interactive customization process to predict user preferences based on their historical interactions. Additionally, expanding the research scope to encompass a broader population can reveal the impact of culture on design choices, thereby enriching the customization framework. Ultimately, our goal is to continuously refine the co-creation service and the aesthetically fitting system, ensuring their alignment with user needs and technological advancements.

6 Conclusion

In conclusion, this project has effectively addressed two **Research Problems** and bridged two corresponding **Knowledge Gaps** in the field of eyewear customization. First, it designs an aesthetically fitting system that aligns eyeglass frames with various face shapes and individual facial features. In this way, it has tackled the challenge of aesthetic coherence in 3D-printed eyewear, thereby enhancing its visual appeal. Second, the development of a co-creation service fosters user engagement and allows consumers to actively participate in the design process, emphasizing the importance of user feedback. Third, we have filled the identified Knowledge Gaps by integrating facial data into the design of customizable eyewear and implementing a co-creation service that enhances personalization—an approach not fully explored in the existing market. This project not only contributes to theoretical frameworks surrounding mass customization and aesthetic design but also lays the groundwork for future advancements in personalized products. Consequently, it paves the way for further integration of advanced technologies, such as artificial intelligence and machine learning, to refine customization processes and improve user experiences.

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